



Verified Carbon Standard

220 MWDC MINBU SOLAR POWER PLANT PROJECT



South Asia

Document Prepared by TÜV SUD South Asia Private Limited

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Summary:

- A description of the project

The project activity involves commissioning and operation of 220 MWDC solar power plant located in Minbu district, Magway region of Myanmar. The project will be commissioned in 4 phases with capacity of 50 MWDC during phase 1, 2 and 3 each and 70 MWDC during phase 4. Currently, only 50 MWDC capacity is commissioned and operational, whereas phase 2 is under construction with expected commissioning in 2024 and phase 3 & 4 in 2025 & 2026 respectively. The project activity exports electricity generated from solar power plant to Electric Power Generation Enterprise (EPGE), thus replacing equivalent amount of electricity generation from fossil-fuel dominant power plants of EPGE. The total net GHG emission reductions generated during the monitoring period of 27-September-2019 to 31-December-2022 is 157,103 ton of CO₂e (tCO₂e).

- The description, purpose and scope of validation and verification

The objective of validation is to assess project conformance to VCS rules, to applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description (PD) will generate verifiable GHG data and information when implemented. The objective of verification is to evaluate the monitoring report and assess the extent to which methods and procedures, including monitoring procedures have been implemented in accordance with the validated project description. This includes ensuring conformance with the monitoring plan. In addition, the extent to which GHG emission reductions report in the monitoring report and materially accurate. This is a joint validation and verification activity and scope includes assessment of PDMR (Project Description and Monitoring Report), ER (Emission Reduction) calculation, and other supporting documents provided during this process against applicable requirements of VCS standard, applied methodology and tools as explained in the section below.

- The method and criteria used for validation and verification

Joint Validation and Verification is conducted using TÜV SÜD South Asia Private Limited procedures in line with the requirements specified in the version 3.2 of the VCS Validation and Verification Manual and ISO 14064-3. The verification team assessed the project activity's compliance against the VCS Standard Version 4.4, VCS Program Guide Version 4.3, the selected methodology ACM0002 Version 21, applied tools. The project is eligible under sectoral scope 1 Energy (renewable/non-renewable).

- The number of findings raised during validation and verification

During joint validation and verification, 3 CARs, 5 CLs and 1 FAR was raised. The assessment is included in the report.

- Any uncertainties associated with the validation and verification

There are no restrictions of uncertainty.

- Summary of the validation and verification conclusions

TÜV SÜD South Asia Private Limited carried out the Verified Carbon Standard (VCS) joint validation and verification of the project, "220 MWDC Minbu Solar Power Plant Project" (4030) with regard to the relevant requirements of VCS Standard Version 4.4. TÜV SÜD confirms all validation and verification activities including objectives, scope and criteria, level of assurance, PDMR adhere to VCS Standard Version 4.4 and all associated updated as documented in this report, are complete. TÜV SÜD concludes that the project activity "220MWDC Minbu Solar Power Plant Project" in Myanmar, as described in the final PDMR version no. 5 dated 06/03/2025, meets all relevant requirements for VCS validation and

verification activity and correctly applied the methodology ACM0002, Version 21. Hence TÜV SÜD can certify that the GHG emission reductions from the project during the monitoring period 27-September-2019 to 31-December-2022 amount to 157,103 tCO₂e.

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1 INTRODUCTION

1.1 Objective

GEP(Myanmar) Co., Ltd. has commissioned the TÜV SUD South Asia Private Limited (TÜV SUD) to carry out the Verified Carbon Standard (VCS) joint validation and verification of the project for the monitoring period from 27-September-2019 to 31-December-2022/VVB Contract/. The objective of validation is to assess project conformance to VCS rules, to applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description (PD) will generate verifiable GHG data and information when implemented. The objective of verification is to evaluate the monitoring report and assess the extent to which methods and procedures, including monitoring procedures have been implemented in accordance with the validated project description. This includes ensuring conformance with the monitoring plan. In addition, the extent to which GHG emission reductions report in the monitoring report and materially accurate. The evaluation is done against the requirements of the VCS Standard, Version 4.4, based on PDMR.

1.2 Scope and Criteria

The scope of validation is to validate that: (a) project description conforms to VCS rules, applied methodology, additionality (b) the project description will generate verifiable GHG data and information. The scope of verification is to verify that: (a) the actual monitoring system and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan; (b) the GHG emission reductions data and express a conclusion with a reasonable level of assurance about whether he reported GHG emission reductions are free from material misstatement; (c) the reported GHG emissions data is sufficiently supported by evidence. Verification shall ensure that the reported GHG removals by sink are complete and accurate in accordance with the applicable VCS criteria to be certified. Validation and verification are conducted using TÜV SUD procedures in line with the requirements specified in the VCS Program and ISO14064-3 requirements and applying auditing techniques. Assessment team assessed and determined that the implementation and operation of the project activity, and steps to report GHG emission reductions comply with the VCS rules. The validation and verification involved a document review of relevant documentation and interview with the project participants. Validation and verification are not meant to provide any consultancy towards the project participants; however, stated request for clarifications and/or corrective actions may have provided input for improvement of the monitoring.

1.3 Reasonableness of Assumptions and Level of Assurance

All GHG sinks, sources and GHG emissions equal to or greater than 5% of the total GHG assertions are considered. As per registered PDMR, the project size is categorized as Projects, regarding its scale, subject to a 5% materiality threshold in line with the section 4.1.8 of the VCS Standard, v4.4 for validation and verification. The level of assurance for verification is reasonable. There are no uncertainties involved in validation and verification.

1.4 Summary Description of the Project

The project activity involves commissioning and operation of 220 MWDC solar power plant located in Minbu district, Magway region of Myanmar. The project will be commissioned in 4 phases with capacity of 50 MWDC during phase 1, 2 and 3 each and 70 MWDC during phase 4. Currently, only 50 MWDC capacity is commissioned and operational, whereas phase 2 is under construction with expected commissioning in 2025 and phase 3 & 4 in 2026 & 2027 respectively. The project activity exports electricity generated from solar power plant to Electric Power Generation Enterprise (EPGE), thus replacing equivalent amount of electricity generation from fossil-fuel dominant power plants of EPGE. The total net GHG emission reductions generated during the monitoring period of 27-September-2019 to 31-December-2022 is 157,103 tCO₂e.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

The objective of validation is to check project conformance according to applicable VCS rules and verification is the ex-post assessment of the monitored GHG data and information. During the validation and verification, the assessment team has evaluated PDMR and assessed the following:

- Project conformance to VCS rules, applied methodology and tools including additionality and extent to which project description can generate verifiable GHG data and information upon implementation.
- The extent to which methods and procedures, including monitoring procedures, have been implemented in accordance with the validated project description. This included ensuring compliance with monitoring plan.
- The extent to which GHG emission reductions reported in the PDMR version 5 dated 06/03/2025 are materially accurate.

In determining the scope of joint validation and verification, TÜV SUD has taken into account the physical boundaries, project sites and temporal boundaries i.e., the length of monitoring period.

A project specific validation and verification audit plan was developed to guide the auditing process to ensure efficiency and effectiveness. The purpose this plan is to present a risk assessment for determining the nature and extent of validation and verification procedures necessary to ensure the risk with respect to material errors, omissions, and misrepresentations is reduced to a reasonable level and there is reasonableness in

assumptions and methods used in validation. According to the ISO14064-3:2019^{/ISO/}, the criteria are the policy, procedure or requirement used as reference against which evidence is compared. Therefore, validation and verification of PDMR was carried out for compliance against the key requirements in the following documents:

- VCS Standard, v4.4^{/VCS/}
- VCS Program Guide, v4.3^{/VCS/}
- VCS Program Definitions, v4.3^{/VCS/}
- ISO 14064-2:2019^{/ISO/}
- ISO 14064-3:2019^{/ISO/}
- Applied Methodology ACM0002 “Grid-connected electricity generation from renewable sources” Version 21^{/Meth/} and Tool01 “Tool for the demonstration and assessment of additionality” Version 7, Tool07 “Tool to calculate the emission factor for an electricity system” Version 7^{/Tool/}.

As per 4.1.2 of VCS Standard, v4.4, the assessment team gathered evidence to verify statement of historical data and information of emission reductions of the monitoring period 27-September- 2019 to 31-December-2022 to a reasonable level of assurance that the project meets threshold of 5% materiality requirement in line with 4.1.8 of VCS Standard, v4.4.

The validation and verification process derived from all items in the criteria stated above. Auditing techniques based on the project parameters, scope, and best professional judgement of the assessment team in order to meet a reasonable level of assurance for validation and verification. The process consisted of the following three phases:

- Document review
- On-site assessment and interviews of project personnel
- The resolution of outstanding issues and the issuance of the final verification report and statement

The assessment team member and technical reviewer composition are listed in the below table:

Table 2.1.1 Assessment team member

	Role		Last name	First name	Affiliation	Involvement in
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N o.					(e.g. name of central or other office of DOE or outsourced entity)	Desk/document review	On-site inspection	Interviews	Verification findings
1.	Team Leader, Validator, Verifier	IR	Kudtarkar	Shruti	TÜV SUD South Asia Pvt. Ltd.	☒	☒	☒	☒
2	Team Leader, Validator, Verifier	IR	Selvaraj	Srinivasan*	TÜV SUD South Asia Pvt. Ltd.	☐	☐	☐	☐
3	Validator, Verifier	IR	Mukherjee	Priyanka*	TÜV SUD South Asia Pvt. Ltd.	☐	☐	☐	☐
4	Financial Expert	EI	Date	Vivek*	TÜV SUD South Asia Pvt. Ltd.	☐	☐	☐	☐

Note: * Srinivasan Selvaraj, Priyanka Mukherjee and Vivek Date were only involved in the preparation of PRR Round 3 responses to VERRA and hence the check boxes are not marked.

Table 2.1.2 Technical reviewer and approver of the verification report

No.	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of DOE or outsourced entity)
1.	Technical reviewer	IR	Raychoudhury	Rishi Kishore	TÜV SUD South Asia Pvt. Ltd.
3.	Approver	IR	Chowdhury	Deepankar	TÜV SUD South Asia Pvt. Ltd.

Key milestones of the joint validation and verification process and corresponding dates are included in the table below:

Milestone	Date
Signing of contract for validation and verification	27/12/2022

Public comment period	20/02/2023 to 22/03/2023
Desk review	15/05/2023
On-site audit	18/07/2023
Issuance of DVR including findings	26/07/2023
Preparation of FVR	12/11/2023
Submission of FVR to VERRA	30/11/2023

2.2 Document Review

The VCS PDMR version 1 dated 13/01/2023^{/MR/}, ER calculations version 1 dated 13/01/2023^{/ER/} and supporting documents related to the project implementation and monitoring were reviewed. Document review was conducted to ensure consistency with and identify any deviation from VCS program requirements. Desk review included an examination of the project details, data and parameters, and quantification of GHG emission reductions.

Furthermore, the assessment team used additional documentation by third parties like host party legislation, technical reports referring to the monitoring or to the basic conditions and technical data. The references used during this verification are summarized in Appendix 2.

The joint validation and verification was performed based on the documents check, site inspection and interviews with relevant personnel. Please refer to section 4 of this report for the process detail and corresponding documents review.

2.3 Interviews

The objective of the interview process was to seek important information from personnel related to project and relevant to the joint validation and verification process. Onsite interviews and information discussions were conducted with the project proponent^{/Insp/}, technical team managing the project operations and maintenance^{/Insp/}, local stakeholders(villagers)^{/Insp/}. The interviews were performed by the assessment team on-site and the following is a list of the main interviewees and subjects discussed.

2.4 Site Visits

The site audit was conducted on 18-July-2023 to verify the actual implementation and operation status of the project activity, monitoring arrangements for monitoring of data and parameters as per PDMR and ACM0002, stakeholder consultation meeting and ongoing stakeholder feedback mechanism, technical details of the solar power plant, logbooks/records maintained at the project site related to operation, maintenance, training, calibration etc. During the site audit, the assessment team was accompanied by the project proponent and site staff.

2.5 Resolution of Findings

Material discrepancies identified during verification are addressed either as CARs, CLs or FARs.

A **Corrective Action Request (CAR)** is established where:

1. The project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions,
2. VCS Standard Version 4.4 requirements have not been met,
3. There is a risk that the emission reductions cannot be monitored or calculated.

A **Clarification Request (CL)** will be issued where information is insufficient, unclear or not transparent enough to establish whether a requirement is met.

A **Forward Action Request (FAR)** will be issued when certain issues related to project implementation should be reviewed during the next verification.

A detailed list of the CARs CLs and FAR raised and discussed during this the joint validation and verification is included in Appendix 3 of this report.

There were 3 CARs, 5 CLs and 1 FAR raised. The assessment is included in the report.

Forward Action Requests

2.5.1

FAR 1- During the joint validation and verification, only 50 MWDC capacity of the total 220 MWDC was commissioned & operational. VVB conducting 2nd verification should check commissioning of remaining capacity of project activity.

3 VALIDATION FINDINGS

3.1 Project Details

- Project type, design technologies and measures implemented, and eligibility of the project

Project involves grid connected 220 MWDC solar power plant supplying electricity generated to EGPE/FS, ESIA, EPC, MIC/. PP has 30 years of Power Purchase Agreement (PPA) with EGPE for sale of electricity/PPA/. Out of 220 MWDC, only 50 MWDC is commissioned and operational/commissioning certificate/. Remaining capacity of 170 MWDC is scheduled to get commissioned in phase wise manner till end of 2027. The solar power plant uses photovoltaic (PV) module of JETION 310-320W/Tech Spec, Equipment list/. Project type is renewable energy generation. Project is eligible for registration under VCS program as it meets requirements of 2.1.1, 2.1.2 and 2.1.3 of VCS Standard 4.4.

- Project proponent and other entities involved in the project

Project proponent GEP (Myanmar) Co. Ltd. There is no other entity involved in the project.

- Ownership

Ownership of the project lies with GEP (Myanmar) Co. Ltd. It is checked from the MIC permit/MIC/ that the investment in the project has been done by GEP (Myanmar) Co. Ltd and

MIC has approved the investment to develop the project. This is in line with 3.7.1 1) of VCS Standard 4.4 wherein project ownership granted under regulation by competent authority.

- Project start date

Project start date is the commissioning date of 50MWDC which is 27-September-2019 as confirmed from the letter issued by Ministry of Electricity and Energy for determining commercial operation date/commissioning certificate/. Start date selected is found to be in line with 3.8 of VCS 4.4. Further, VERRA has granted the project exemption from section 3.7.1 to complete validation by November 2023/Exemption/.

- Project crediting period

PP has chosen 7 years of crediting period which is twice renewable. The first crediting period is from 27-September-2019 to 26-September-2026. The total crediting period is from 27-September-2019 to 26-September-2040. This is in line with 3.9.1 of VCS Standard 4.4.

- Project scale and estimated GHG emission reductions or removals

Estimate average GHG emission reductions per annum is 155,248 tCO_{2e}. Project scale is “Project” as emission reductions are less than 300,000 tCO_{2e} per year.

- Project location

Project is located in Minbu district, Magway Region, Myanmar. The physical coordinates are 20° 02'57.9" N 94° 40'58.1E". The location and physical coordinates were checked during on-site audit.

- Conditions prior to project initiation

Project is a greenfield project as confirmed from feasibility study, ESIA report/^{FS, ESIA/} and MIC permit/^{MIC/}. Prior to project initiation, equivalent amount of electricity would have been produced in fossil-fuel dominant grid of Myanmar. The same scenario has also been identified as the baseline scenario by PP which is deemed to be acceptable.

- Project compliance with applicable laws, statutes and other regulatory frameworks

In the PDMR, PP has indicated national laws which are applicable to the project activity and it has obtained desired permits/license from competent authority under these laws/^{MIC, RC, EPC, ESIA/}. These laws are as follows:

1. The Electricity Law 2014
2. The Environmental Conservation Law 2012
3. The Forest Law 1992
4. Myanmar Investment Law 2016

5. Minimum Wage Law 2013
6. The Settlement of Labour Dispute Law 2012
7. The Social Security Law 2012

Based on the review of permits/license project has obtained, Project's compliance with applicable national laws and regulations have been found to be in order.

- Participation under other GHG programs:

The GHG emission reductions generated by the project have not been counted or used under other GHG Programs i.e. UNFCCC-CDM, The Gold Standard (GS), Global Carbon Council (GCC) and/or other forms of environmental credit known to the assessment team. The net GHG emission reductions from the project will not be used for compliance with emission trading programs or to meet binding limits on GHG emissions. The host country Myanmar does not have a national emissions trading scheme. Thus, it has been concluded that the project is not part of any emission trading program or it has not participated under any GHG programs and the same has been confirmed via checking the UNFCCC-CDM/UNFCCC-CDM/, GS /GS/, GCC/GCC/ and confirmation from the project proponent/declaration/.

- Other forms of credit and supply chain (Scope 3) emissions:

The project is not part of emission trading programs and other binding limits as declared by the PP/declaration/. Further, it does not seek any other form of environmental credit/declaration/. Project is not part of scope 3 supply chain.

- Additional information relevant to the project, including:
 - Sustainable development contributions

In addition to SDG (Sustainable Development Goal) 13 take urgent action to combat climate change and its impacts, project has contributed to SDG 7 affordable and clean energy and SDG 8 decent work and economic growth. Project's contribution to these SDGs has been checked by the assessment team based on documentation review such as ER calculation/ER/ for SDG 13, invoices raised to EGPE/INV/ for SDG 7 and labour charge form/Daily Wage record/ for SDG 8 and it has been found to be satisfying the performance indicators chosen under these SDGs.

Overall, the description in the PDMR is accurate, complete, and provides an understanding of the nature of the project, and whether the project has been implemented as described in the project description. The project is likely to achieve estimated GHG emission reductions of 1,086,738 tCO₂e in the first crediting period, the actual results may vary since the estimates are based on assumptions that are subject to change.

3.2 Participation under Other GHG Programs

The GHG emission reductions generated by the project have not been counted or used under other GHG Programs i.e. UNFCCC-CDM, The Gold Standard (GS), Global Carbon Council (GCC)

and/or other forms of environmental credit known to the assessment team. The net GHG emission reductions from the project will not be used for compliance with emission trading programs or to meet binding limits on GHG emissions. The host country Myanmar does not have a national emissions trading scheme. Thus, it has been concluded that the project is not part of any emission trading program or it has not participated under any GHG programs and the same has been confirmed via checking the UNFCCC-CDM/^{UNFCCC-CDM/}, GS/^{GS/}, GCC/^{GCC/} and confirmation from the project proponent/^{declaration/}.

3.3 Safeguards

No Net Harm

3.3.1

As per EIA procedure 2015, monitoring report is required to be submitted every 6 months to Environmental Conservation Department (ECD) showing compliance with Environmental Management Plan (EMP). The negative environmental and socio-economic impacts have been mitigated by taking necessary actions details of which are verified from the biannual monitoring reports from Jan 2018 to Dec 2022 prepared by Myanmar Survey Research/^{Biannual Report/}. Assessment team has provided only snippet of detailed monitoring report to elucidate how potential negative impacts are being mitigated:

1. Construction activity causes dust and emissions from exhaust of construction equipment. Dust emissions are prevented or reduced by implementing measures such as sprinkling water on soil before excavation, avoid pouring dust materials from elevated areas to ground and exhaust emissions by controlling speed and operation of construction activities, vehicle idling minimization, regular maintenance of equipment etc.
2. Noise emanating from construction machinery. The impact of this is prevented by using machinery with noise suppressing technology, installing portable barriers to shield compressors & stationary equipment to reduce noise, use of PPEs, regular maintenance of machinery etc.
3. To mitigate other negative environmental/social impacts such as hazardous waste generation from solar panels at the end of life of plant, impact on land, ecosystem, PP has implemented EMP. Bi-annual monitoring reports cover detailed assessment of projects compliance with EMP and have been reviewed to confirm that PP has implemented mitigatory actions to reduce negative environmental, socio-economic, health impacts.

3.3.2

Local Stakeholder Consultation

The PP has conducted local stakeholder consultation on 28 September 2022. The local authority at Lay Pin Village was intimated regarding this. The meeting was to inform villagers about issuance of carbon credits with VERRA and its associated benefits to stakeholder. Local stakeholder consultation records/^{lsc/} were checked to confirm the communication letter sent to local authority, Minutes of meeting of the consultation and found to be in order.

There were no negative feedback about the project and consultation inputs received from stakeholders did not require change in the project design.

Ongoing communication with stakeholders is maintained through calls, email or in person/^{ESIA report/}. There were no complaints or negative feedback received from stakeholders in the ongoing communications. Overall, the local stakeholder consultation was found to be in order with VCS requirements.

Environmental Impact

3.3.3

ESIA report/^{ESIA/} has been prepared by Myanmar Survey Research dated 20 October 2016 to identify and discuss implications of environmental impact assessment and environmental management plan (EMP) to mitigate negative impacts. Assessment team has verified ESIA report to confirm that the report presents in depth study of the project's impact of environment and provides in detail view of EMP. Some of the environmental impacts identified by MSA during construction include soil erosion, dust emissions, air pollution, noise pollution, water usage, oil spill, sewage, solid waste and construction waste etc. Annex 2 of the ESIA report provides EMP for construction and operation phase and budget allocated to implement mitigatory actions and monitoring frequency for environmental impacts.

3.3.4

Public Comments

PDMR was available for public comments on VERRA website during 20/02/2023 to 22/03/2023. No comments were received during this period/^{VERRA website/}.

3.3.5

AFOLU-Specific Safeguards

This section is not applicable as project is non-AFOLU.

3.4.13.4

Application of Methodology

Title and Reference

Methodology applied is:

ACM0002 Grid-connected electricity generation from renewable sources Version 21

Tools applied in relation to the methodology are:

Tool01 "Tool for the demonstration and assessment of additionality" Version 7

Tool07 "Tool to calculate the emission factor for an electricity system" Version 7

Tool27 "Investment analysis" Version 13

The versions of methodology and tools applied are latest available at the time of validation and verification.

Applicability

Assessment of applicability of the methodology, and any tools is discussed below:

Table 3.4.2.1 Assessment of ACM0002 Version 21

3.4.2

Applicability condition of ACM0002	Assessment of fulfilment of applicability condition
This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	Assessment team has confirmed by verifying permit issued by MIC/^{MIC}/ that the project activity is a greenfield solar power plant. Hence, condition a is fulfilled.
In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	Assessment team has confirmed from MIC permit/^{MIC}/, commissioning certificate/^{commissioning certificate}/ and verification done during on-site audit/^{Insp}/ that the project activity does not involve integration of BESS.
The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity	As discussed in the assessment of above 2 conditions, this condition is not applicable.

addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies² may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
In the case of integrated hydro power projects, project participants shall: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density,	Project is a grid-connected solar power plant. Hence, this condition is not applicable.

calculated using equation (7), is greater than 4 W/m²; or — (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; — (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account. In the case of integrated hydro power projects, project participants shall: seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	Project is a grid-connected solar power plant. Hence, this condition is not applicable.

The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site;	Project is a grid-connected solar power plant. Hence, this condition is not applicable.
(b) Biomass fired power plants/units.	
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Assessment team has confirmed by verifying permit issued by MIC/MIC/ that the project activity is a greenfield solar power plant. Hence, this condition is not applicable.
In addition, the applicability conditions included in the tools referred to below apply.	Applicability of the applied tools has been verified and discussed in the tables below.

Table 3.4.2.2 Assessment of Tool01

Applicability condition of Tool01	Assessment of fulfilment of applicability condition
The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Assessment team has confirmed that PP uses an approved existing methodology ACM0002. Hence, this condition is not applicable.
Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Assessment team has confirmed that 5.3.2 of ACM0002 Version 21 refers to additionality demonstration using Tool01. Hence, this condition is met.

Table 3.4.2.3 Assessment of Tool07

Applicability condition of Tool07	Assessment of fulfilment of applicability condition
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or	Assessment team has confirmed that emission factor calculation takes into consideration OM, BM and CM when

a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	calculating baseline emissions/ ^{ER} , ^{PDMR} /. Hence, this condition is not applicable.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option II a and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Assessment team has confirmed emission factor calculation is based on grid connected power plants only/ ^{ER} , ^{PDMR} /. Hence, this condition is met.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Project exports electricity to EGPE, Myanmar which is non-Annex I country. Hence, this condition is not applicable.
Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	Not applicable as project is grid-connected solar power plant.

Table 3.4.2.4 Assessment of Tool27

Applicability condition of Tool27	Assessment of fulfilment of applicability condition
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Assessment team has confirmed PP has applied and demonstrated applicability to Tool07/ ^{PDMR} /. Hence, this condition is met.

In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	ACM0002 doesn't contain specific requirements for investment analysis. Hence, this condition is not applicable.
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In conclusion, assessment team has ascertained that the project meets relevant applicability conditions of applied methodology and tools.

Project Boundary

In accordance with 5.1 of ACM0002 version 21, PP has chosen project boundary to include 220MWDC solar power plant and EGPE grid to which the project exports power to and in turn power plants connected to EGPE grid. The spatial extent of project boundary was found to be appropriate.

As per 23 of methodology, PP has selected emission sources and gases indicated in the table. Assessment of this is provided in the table below.

Table 3.4.3.1 Assessment of emission sources and gases included or excluded from the project boundary

Emission Source	Gas	Included or excluded	Assessment opinion
CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO ₂	Included	This is the main emission source and gas. Hence, inclusion is appropriate.
	CH ₄	Excluded	This is minor emission gas. Exclusion is allowed by methodology. Hence is appropriate.
	N ₂ O	Excluded	This is minor emission gas. Exclusion is allowed by methodology. Hence appropriate.
For dry or flash steam geothermal power plants, emissions of CH ₄ and CO ₂ from non-condensable gases contained in geothermal steam	CO ₂ , CH ₄ , N ₂ O	Excluded	Emission source and gases are excluded as project is not a geothermal power plant. Hence appropriate.
For binary geothermal power plants, fugitive emissions of CH ₄ and	CO ₂ , CH ₄ , N ₂ O	Excluded	Emission source and gases are excluded as project is not a geothermal

CO ₂ from non-condensable gases contained in geothermal steam			power plant. Hence appropriate.
For binary geothermal power plants, fugitive emissions of hydrocarbons such as n-butane and isopentane (working fluid) contained in the heat exchangers	CO ₂ , CH ₄ , N ₂ O	Excluded	Emission source and gases are excluded as project is not a geothermal power plant. Hence appropriate.
CO ₂ emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants	CO ₂ , CH ₄ , N ₂ O	Excluded	Emission source and gases are excluded as project is not a solar thermal or geothermal power plant. Hence, appropriate.
For hydro power plants, emissions of CH ₄ from the reservoir	CO ₂ , CH ₄ , N ₂ O	Excluded	Emission source and gases are excluded as project is not a hydro plant. Hence, appropriate.
Charging of BESS using electricity from the grid or from fossil fuel electricity generators	CO ₂ , CH ₄ , N ₂ O	Excluded	Emission source and gases are excluded as project doesn't include charging of BESS. Hence, appropriate.

3.4.4

In overall conclusion, project boundary and selected GHG sources, sinks are justified for the project.

Baseline Scenario

3.4.5

As verified from permit by MIC/MIC/ and on-site audit/Insp/, the project is greenfield solar power plant. Hence, as per 5.2.1 of ACM0002 version 21, baseline scenario for greenfield power plant is electricity delivered to the grid by project activity that would have been otherwise been generated by the operation of grid-connected power plants and by addition of new generation sources, as reflected in CM calculation as described in Tool07. In conclusion, baseline scenario identified was found to be justified.

Additionality

As per 5.3.2 of ACM0002, the additionality of the project activity has been demonstrated and assessed using Tool01 “Tool for the demonstration and assessment of additionality” Version 7.

Regulatory surplus- Project has been implemented in accordance with laws and regulations in the Myanmar as presented in the earlier sections. It was checked by the assessment team that none of these regulations require project activity to be implemented.

Assessment of additionality in line with Tool01 has been presented below:

Step 0: Demonstration whether the project activity is first of its kind

The project is first-of-its-kind project in Myanmar as confirmed by the assessment team based on the publicly available information. Following references clearly demonstrate the first of its kind:

1. *“Minbu 220MWDC Solar Power Plant” Phase 1 (50 MWDC) Grand Opening Honoured by Aung San Suu Kyi (bangkokpost.com)*
2. *State Counsellor launches first phase of solar power project in Minbu | Myanmar Digital News (mdn.gov.mm)*
3. *Southeast Asia’s ‘largest solar power plant’ to be built at Minbu | Frontier Myanmar*
4. *Myanmar’s First Solar Power Plant Joins National Grid (irrawaddy.com)*
5. *Myanmar Opens First Solar Power Plant | Greater Mekong Subregion (GMS)*
6. *First commercial solar power plant inaugurated in Myanmar central region - Xinhua | English.news.cn (xinhuanet.com)*
7. *First commercial solar power plant launched in Myanmar (consult-myanmar.com)*
8. *Myanmar launches first solar power plant - Open Development Mekong*

Information available in the public domain as verified from the above references demonstrate that the project activity is first of its kind in Myanmar. Assessment team has checked for registered or under validation or requesting registration projects under various carbon offset programs eg. UNFCCC-CDM, VCS, GS, GCC, CAR. The verification didn’t reveal any grid-connected solar power plant project situated in Myanmar. However, PP has further gone to demonstrate the additionality using investment analysis since first-of-its kind approach limits the crediting period to 7 years (renewable twice).

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Step 1a: Define alternatives to the project activity

Alternative available to the project participant or similar project developers that provide outputs or services comparable with the project activity is identified below:

Alternative (a): Continuation of the current situation (no project activity or other alternative undertaken).

In case that project activity is not undertaken, the grid would still continue with the fossil fuel based power projects and this would result in GHG emissions.

Step 1b: Consistency with mandatory laws and regulations.

The project activity complies with national policies for environmental protection, energy conservation, and sustainable development, and there are no binding legal and regulatory requirements for this project type. Furthermore, by exceeding the regulatory standards for renewable energy production, the project has the potential to create a regulatory surplus. This surplus, in the form of additional carbon credits, can be traded or sold, providing financial incentives for further environmental improvements and reinforcing the project's alignment with national and international sustainability goals.

Step 2: Investment analysis

Sub-step 2a: Determine appropriate analysis method

Tool for the Demonstration and Assessment of Additionality (version 07.0) recommends three analysis methods, namely simple cost analysis (Option I), investment comparison analysis (Option II) and the benchmark analysis (Option III). The project is the first solar power plant in Myanmar so alternative comparison cannot be applied. Option I and option II is not appropriate. The benchmark analysis method (Option III) is the most appropriate option. It will be analysed in the Sub-step 2b below.

Sub-step 2b: Apply benchmark analysis

According to the Methodological Tool for Investment Analysis, Version 13.0, Section 6, paragraph 15, "Local commercial lending rates or weighted average costs of capital (WACC) are appropriate benchmarks for a project IRR. The VVB noted that the PP has used Post-tax equity IRR for investment analysis, benchmarking against the cost of equity. According to the Methodological Tool for Investment Analysis, Version 13.0, Section 6, paragraph 19, using option a, the default value of the cost of equity in the appendix for Myanmar is used as the benchmark.

The VVB noted that Equity IRR calculations use nominal input values from the FSR, while Myanmar's default cost of equity given in the appendix of Tool 27 is in real terms. According to Tool 27, paragraph 16, this should be adjusted by adding the inflation rate. However, no inflation forecast or target rate from the Central Bank of Myanmar is available. The VVB noted that IMF provides inflation rates for Myanmar from 2019 to 2023 as follows: 8.6%, 5.7%, 3.6%, 18.4%, and 27.1%. If the default cost of equity (20.98%) is adjusted with the average inflation rate (12.68%), the real term value is 33.66%. The PP has used the default value of 20.98% without adjustment, which is found to be conservative.

Given the period of time between the Feasibility Study Report and the COD of the first phase, the

audit team was therefore confident that it is unlikely in the context of the underlying Project that the input values would have materially changed. At the same time, the audit team compared the input values for the financial analysis in the PD and the Feasibility Study Report and confirms that all input parameters used in the financial analysis are taken from the Feasibility Study Report. Therefore, the audit team confirms that the input values from the Feasibility Study Report were valid and applicable to the investment analysis. VVB assessed that According to MIC Approval in no.6 9(a) under Section 27(a), the income tax exemption is granted for five consecutive years, starting from the year of commencement of commercial production. The IRR assumptions indicate that commercial production begins in 2018, and therefore, the MIC tax holiday applies from 2018 to 2022. This aligns with the actual accounting practices of MIC, where the exemption is applied for the first five years following the project's commissioning in 2019.

It was also noted that aside from the MIC tax holiday, the project does not receive any other tax benefits, subsidies, or accelerated depreciation, which was confirmed by VVB by verifying the Publication Titled “Renewable Energy Snapshot” Published in 2019 /RES/. The PP has included this tax exemption in the PD input values.

VVB has verified the Revised PDMR and revised IRR excel file and found the calculations to be appropriate.

Sub-step 2c: Calculation and comparison of financial indicators

For the calculation of the financial indicators for the project, the parameters listed in Table 1 are applied. As per the CDM tool-27 “Investment Analysis”, which is in accordance with the Feasibility Study Report (FSR).

The VVB noted that the feasibility Study Report of the project was designed by independent third party, by Shinawatra University of the Kingdom of Thailand with the supervision of Dr. Pruitipong Thaicham, Professor of Renewable Energy., which has been authorized design qualification. Hence, VVB concluded that the values in Feasibility Study Report can be considered independent, appropriate and realistic.

Table 3.4.5.1: Main assumptions parameters

Item	Value	Unit	Source
Revenue Assumptions			
Power Plant Net Capacity (MW)	220	MW	FSR page 86
Phase 1	50	MW	FSR page 86
Phase 2	50	MW	FSR page 86
Phase 3	50	MW	FSR page 86
Phase 4	70	MW	FSR page 86

COD date			
Phase 1	Apr-18	Date	FSR page 86, 88
Phase 2	Apr-19	Date	FSR page 86, 88
Phase 3	Apr-20	Date	FSR page 86, 88
Phase 4	Apr-21	Date	FSR page 86, 88
End date			
Phase 1	Mar-48	Date	FSR page 86
Phase 2	Mar-48	Date	FSR page 86
Phase 3	Mar-48	Date	FSR page 86
Phase 4	Mar-48	Date	FSR page 86
Energy Production yield	1,614.00	kWh/MWp/year	FSR page 86
Electricity tariff	0.1275	USD/KWh	FSR page 86
Degradation Rate for year 2 nd onward	0.70%	%	FSR page 52
Project Operation Life Cycle (years)	30	year	FSR page 86
Cost Assumptions			
Administration expense	500,000	Base price in USD	FSR page 86
Administration expense increase	10%	% increase in every 3 years	FSR page 88
O&M Cost	3,000,000	Base price in USD	FSR page 86
O&M expense increase	10%	% increase in every 10 years	FSR page 88
CSR (of net Income)	2%	% of Net Income	FSR page 86
Land lease	2,390,000	USD paid in 1 st year	FSR page 89
Insurance Premium			
Project Risk	0.15%	of total investment	FSR page 86

Business Interruption	0.15%	of debt service	FSR page 86
Third party Liability	20,000	per year	FSR page 86
Political Risk	2.40%	of debt service	FSR page 86
Tax Assumptions			
Income Tax	25%	%	FSR page 87
MIC Tax Holiday	5	Years from commissioning of project	FSR page 87
Commercial Tax	5.00%	year	FSR page 87
Financing Assumptions			
Loan rate	65%	%	FSR page 87
Equity	35%	%	FSR page 87
Interest Rate	7.47%	%	FSR page 87
Loan Period	15	years	FSR page 87
Grace Period	-	years	FSR page 87
Condition of Repayment	2.50%	each year during 1 st -3 rd year	FSR page 87
	5.00%	each year during 4 th -6 th year	FSR page 87
	7.50%	each year during 7 th -12 th year	FSR page 87
	10.00%	each year during 13 th -14 th year	FSR page 87
	12.50%	each year during 15 th year	FSR page 87

Equity Arrange Fee	0.00%	of total Equity	FSR page 87
Loan Arrange Fee	1.50%	of total Loan	FSR page 87
CAPEX			
Phase 1	72,770,000	USD	FSR page 85
Phase 2	66,890,000	USD	FSR page 85
Phase 3	63,880,000	USD	FSR page 85
Phase 4	89,070,000	USD	FSR page 85
Total CAPEX	292,610,000	USD	FSR page 85
Residual value %	10%	%	https://www.nrel.gov/docs/fy19osti/72217.pdf
Residual value	29,261,000	USD	Calculated
Net depreciable value	263,349,000	USD	Calculated
Loan	190,196,500	USD	FSR page 87
Equity	102,413,500	USD	FSR page 87

The values for the input parameters are based on FSR The project has offered by EPGE with agreed tariff of USD 12.75 cents for the project capacity of 220 MW DC for a period of 30 years. Energy production yield and degradation rate of the solar panel is based on solar panel that the project used (FSR page 89 and 52 respectively). Project cost is based on estimated total costs in FSR. The project cost is depreciated using the straight-line method, and a residual value (RV) of 10% of the total CAPEX is added back to the cash flow in the final year, aligning with the total project life cycle. Due to the absence of specific accounting regulations in Myanmar regarding RV calculation, the 10% value of the total CAPEX is derived from a technical report published by NREL¹. The land used by the project is a leased land from Government and hence the PP has not added the land cost at the end of the project life time, which is found to be appropriate.

Assessment team has checked NREL technical paper and concluded that 10% of residual value is considerably conservative. Also residual value which is the remaining value of an asset after it has been fully depreciated, or after deteriorating beyond further use and has been added back as cash inflow.

¹ <https://www.nrel.gov/docs/fy19osti/72217.pdf>

The IRR calculation process in the IRR calculation spreadsheet has been checked and confirmed in accordance with the CDM Tool 27.

The result of the calculation is shown in table below.

Table 3.4.5.2: Financial indicator comparison

IRR without ER revenue	Benchmark
9.55 %.	20.98%

By using the main assumptions stated in Table 3.4.5.1, proposed equity IRR (post tax) of the proposed project is 9.55 % which is lower than the benchmark of 20.98% which means that the project is not financially attractive.

Assessment team has checked all the assumption and noted that they have been sourced from Feasibility Study report and found consistent.

Step 2d: Sensitivity analysis

The VVB has assessed the sensitivity analysis in accordance with the requirements outlined in CDM Tool 27 (Investment Analysis). The assessment evaluates the appropriateness of the sensitivity analysis parameters, the validity of assumptions made, and the robustness of the financial conclusions. The key elements of the assessment are as follows:

CDM Tool 27 requires that any variables constituting more than 20% of either total project costs or revenues be subjected to reasonable variation. The PP has conducted sensitivity analysis on the following four financial indicators:

- 1) Electricity tariff
- 2) Annual electricity yield
- 3) Total static investment
- 4) Annual O&M cost

The variation range applied by the PP includes $\pm 5\%$ and $\pm 10\%$, which is acceptable; The results of the sensitivity analysis is provided below.

Variation %	-10%	-5%	5%	10%
Tariff	7.54%	8.54%	10.57%	11.57%
Energy Production yield	7.54%	8.54%	10.57%	11.57%

Investment (Equity)	10.40%	9.96%	9.18%	8.82%
O&M	9.74%	9.65%	9.46%	9.37%

Electricity Tariffs

The PP states that the electricity tariff must increase by 57% to cross the benchmark of 20.98%. However, the tariff is fixed under the Power Purchase Agreement (PPA) at USD 0.1275/kWh, which reduces the likelihood of fluctuation. Given the contractual nature of the tariff, the VVB finds the PP's justification reasonable, as the possibility of a 57% increase in tariff is unrealistic within the project's operational framework.

Annual Electricity Yields

The PP indicates that an increase of 57% in annual electricity yield would be required to surpass the benchmark. The electricity yield was estimated using modeling programs, which accounts for environmental and technical constraints. Since electricity generation is primarily dependent on environmental factors such as irradiance, which cannot be controlled by the project, the assumption that no significant increase beyond the model's prediction is feasible is deemed reasonable by the VVB.

Total Static Investment

The PP states that equity must be reduced by 68% to exceed the financial benchmark. The project has been financed with 35% equity and 65% debt, which is a typical financial structure for such projects. The PP asserts that a reduction in equity to such an extent is not possible given the fixed nature of the investment structure. The VVB finds this reasoning acceptable, as significant changes in financing structures post-implementation are generally impractical.

Annual O&M Cost

The PP has indicated that O&M costs would need to be reduced by 462% for the project to surpass the benchmark, which is an unrealistic assumption. The VVB concurs that O&M costs have an inherent floor due to operational necessities and market rates for maintenance services. Given that negative cost reductions are not feasible, the assumption is valid, and no further sensitivity adjustments are necessary.

The primary objective of the sensitivity analysis is to demonstrate that the financial/economic attractiveness conclusion remains robust under reasonable variations in key parameters. The VVB confirms that:

- The PP has selected appropriate key financial indicators for sensitivity testing.

- The required variation for these indicators to surpass the financial benchmark is significantly beyond plausible real-world fluctuations, as shown in the Table above.
- The assumptions made regarding tariff stability, electricity yield constraints, investment structure rigidity, and O&M cost floors are reasonable and aligned with industry norms.
- The project remains financially unattractive under all reasonable variations, supporting the additionality claim.

CDM Tool 27 mandates that sensitivity analysis results must be reproducible using the provided spreadsheets. The PP has ensured that the financial model, along with calculations, is transparently documented and Sensitivity analysis data is included in the PDD. The VVB can verify and validate the results by cross-checking assumptions, financial formulas, and referenced datasets. Hence, the VVB finds that the PP has conducted a sensitivity analysis in line with the CDM Tool 27 requirements. The assumptions made are reasonable, and the variations applied sufficiently demonstrate that the project remains financially unattractive under all feasible scenarios.

Step 3: Barrier Analysis

The proposed project does not employ the barrier analysis. Then proceed to Step 4.

Step 4: Common Practice Analysis

According to Tool for the demonstration and assessment of additionality, Common practice Version 03.1 is used in this step.

Sub-step 4.1. Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the project activity

The capacity of the project is 220 MWp. Therefore, the projects in the range of 110 MWp~330 MWp should be chosen for this analysis.

Sub-step 4.2. Identify similar projects (both CDM and non-CDM) which fulfil all of the

following conditions:

- (a) The projects are located in the applicable geographical area;
- (b) The projects apply the same measure as the project activity;
- (c) The projects use the same energy source/fuel and feedstock as the project activity, if a technology switch measure is implemented by the project activity;
- (d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the project plant;
- (e) The capacity or output of the projects is within the applicable capacity or output range calculated in Sub-Step 4.2;

(f) The projects started commercial operation before the PD is published for global stakeholder consultation or before the start date of project activity, whichever is earlier for the project activity.

In this case, the solar PV projects with 110 MWp~330 MWp, which have started commercial operation before 27/09/2019 (Letter of commercial operation date) in Myanmar, will be chosen for this analysis. There is no similar project which meets the criteria above.

Sub-step 4.3. within the projects identified in Sub-Step 4.2, identify those that are neither registered CDM (VCS, GS, GCC, CCER) project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number Nall

There is no similar project which meets the criteria above. Therefore, the Nall=0.

Sub-step 4.4. Within similar projects identified in Sub-Step 4.3, identify those that apply technologies that are different to the technology applied in the project activity. Note their number Ndiff.

Nallf=0, Ndiff=0,

Therefore, $F = 1 - N_{diff} / N_{all} = 0 < 0.2$;

Nall- Ndiff=0<3

Outcome of step 4:

From the above analysis and discussion, it is concluded that the project is not common practice. Therefore, the project activity is deemed additional.

3.4.6

Quantification of GHG Emission Reductions and Removals

Assessment team has described below the steps taken to validate the quantification methods, including all data and parameters used in the equations, and any references to any other data sources used.

- Quantification of baseline emissions

Baseline emissions are calculated in line with para 47 of ACM0002 version 21. Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to project activity and calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

EG_{PJ,y} = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using Tool07 (tCO₂/MWh)

As per para 49 of ACM0002, for greenfield power plants, $EG_{PJ,y}$ is $EG_{facility,y}$.

For calculating estimated BE_y , PP has calculated $EG_{PJ,y}$ based on the installed capacity in AC (220 MWDC converts to 170 MWAC) considering phase wise commissioning sourced from PPA^{ER, PPA/}.

For calculating BE_y during monitoring period, PP has sourced from the invoices issued to $EGPE^{Inv/}$.

PP has adopted similar approach for calculating $EF_{grid,CM,y}$ for estimated and actual ER. Since Government of Myanmar has not released combined margin emission factor for its national grid nor there is information available on operating margin and build margin to calculate combined margin in public domain for Myanmar, $EF_{grid,CM,y}$ has been sourced from publicly available IFI dataset for default grid factors version 3.2 published in April 2022^{/IFI/} issued by UNFCCC.

Thus, estimated BE_y for the crediting period in a calendar year is as below:

Year	$EG_{facility,y}$ (MWh)	$EF_{grid,CM,y}$ (tCO ₂ /MWh)	BE_y (tCO ₂ e)
Year 1	80,700	0.602	48,581
Year 2	140,660	0.602	84,677
Year 3	220,230	0.602	132,578
Year 4	323,445	0.602	194,714
Year 5	349,204	0.602	210,221
Year 6	346,719	0.602	208,725
Year 7	344,254	0.602	207,241

Thus, actual BE_y for current monitoring period in a calendar year is as below:

Year	$EG_{facility,y}$ (MWh)	$EF_{grid,CM,y}$ (tCO ₂ /MWh)	BE_y (tCO ₂ e)
27-September-2019 to 31-December-2019	21,411	0.602	12,890
2020	80,934	0.602	48,722
2021	77,461	0.602	46,632
2022	81,162	0.602	48,860

- Quantification of project emissions

As per para 35 of ACM0002, for most renewable energy power generation project activities, project emissions are 0. Further, it has been confirmed during the on-site audit that the project does not involve use of fossil fuel on-site. Hence, PE_y is 0.

- Quantification of leakage emissions

As per para 61 of ACM0002, leakage emissions is 0.

- Summary of net GHG emission reductions

Estimated ERs during the crediting period of 7 years from the project start date are presented in the table below:

Year	BE _y (tCO ₂ e)	PE _y (tCO ₂ e)	L _y (tCO ₂ e)	ER _y (tCO ₂ e)
Year1	80,700	0	0	48,581
Year 2	140,660	0	0	84,677
Year 3	220,230	0	0	132,578
Year 4	323,445	0	0	194,714
Year 5	349,204	0	0	210,221
Year 6	346,719	0	0	208,725
Year 7	344,254	0	0	207,241
Total	1,805,212	0	0	1,086,738

Actual ERs achieved during the monitoring period 27-September-2019 to 31-December-2022 is 157,103 tCO₂e which is presented in the table below:

Year	BE _y (tCO ₂ e)	PE _y (tCO ₂ e)	L _y (tCO ₂ e)	ER _y (tCO ₂ e)
27-Sept-2019 to 31-Dec 2019	12,890	0	0	12,890
01-Jan-2020 to 31-Dec-2020	48,722	0	0	48,722
01-Jan-2021 to 31-Dec-2021	46,632	0	0	46,632
01-Jan-2022 to 31-Dec-2022	48,860	0	0	48,860

- Uncertainties associated with the calculation of emissions

There are no uncertainties associated with the calculation of emissions.

Assessment team has verified following with respect to the project description:

- All relevant assumptions and data are listed in the project description, including their references and sources.
- All data and parameter values used in the project description are considered reasonable in the context of the project.
- All estimates of the baseline emissions can be replicated using the data and parameter values provided in the project description.

In conclusion, ACM0002 version 21 and tool07 have been applied correctly to calculate baseline emissions, project emissions, leakage and net GHG emission reductions during the project crediting period.

Methodology Deviations

3.4.7

Based on the public data in Myanmar, either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, lead to the combined margin emission factor (EF_{grid,CM,y}) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Myanmar.

By checking the UNFCCC website, for the purpose of promoting greater harmonization, the IFI Technical Working Group (IFI TWG) on GHG accounting maintains a common dataset containing Default Emissions Factor (DEF) of the country’s electricity grid including in-country interconnected grids. The DEFs apply to electricity generation in a country and currently do not consider the impact of interconnections with neighbouring countries. According to the public “Methodological Approach for the Common Default Grid Emission Factor Dataset”⁸, the common dataset containing DEFs is constructed using a Combined Margin (CM) for the grid that is calculated by combining an Operating Margin (OM) and a Build Margin (BM)

as per CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0). This calculation method of CM is in line with ACM0002 (Version 21.0). Besides, the country specific information that is used to calculate OM and BM is from The International Energy Agency’s (IEA) energy statistics database. Therefore, the default emission factor from IFI Dataset is accurate and authoritative.

According to the latest IFI Dataset of Default Grid Factors v.3.2 published in April 2022⁸, the latest EF_{grid,CM,y} of solar project and firm energy project (e.g. Hydro, geothermal) in Myanmar in 2021 is 0.602 tCO₂e/MWh, respectively. To be conservative, 0.602 tCO₂e/MWh can be used to calculate emission reduction for the project.

3.4.8

The deviation has no impact on conservativeness of the quantification of GHG emission reductions or removals. The deviation relates only to the criteria and procedures for monitoring or measurement and does not relate to any other part of the methodology.

Monitoring Plan

Monitoring team for the project include following staff:

- Plant director: Responsible for approving the number and all data information of the Power plant.
- Deputy plant manager: Responsible for checking the number, coordinate with relevant Government (EPGE) and information.
- Operation manager: Responsible for collecting data and preparing the right data, operational reports, recording the daily operation of the solar power plant, including operating periods, equipment defects, etc. of the Power plant for other team members.
- Financial Manager: To be responsible for collection of sales receipts.

- EPGE representative (Mann sub-station plant manager): Responsible to cross check the number and approve the number

Electricity import and export is continuously measured using 230kV main and back up meters installed on the feeder to Mann and Ann substation. Data is recorded monthly by site staff and substation representative based on which monthly invoice is raised by PP to EGPE for sale of electricity.

If the error of the main meter exceeds the allowable tolerance or its malfunction occurs, the grid-connected electricity generated by the Power plant will be resolved by following measures:

1. Adopting the backup meter's data (after taking the line losses into consideration), unless a test by either party reveals it is inaccuracy;
2. If the inaccuracy of the backup meter is not within the acceptable limits or it cannot work properly, the project owner and the grid company shall jointly prepare a new agreement of correct reading;
3. If the power plant owner and the grid system fail to reach an agreement concerning the correct reading, this matter will be submitted for arbitration according to agreed procedures.

Meters are 0.5s class and calibrated yearly by EGPE as stated in the PPA. Project has SCADA system to monitor operation of power plant including total generation, export and import of electricity. All data is kept for at least 2 years after the end of crediting period. Site staff includes engineers and experts who are trained by PP regularly on operation, maintenance, quality control, data monitoring, recording and other key aspects of project.

In conclusion, adherence of the monitoring plan to the requirements of the ACM0002 version 21.

3.5 Non-Permanence Risk Analysis

This section is not applicable as project is a non-AFOLU.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

The following table describe for each parameter which data are fixed ex-ante and monitored ex-post. Data and parameters fixed ex-ante and ex-post as listed in the monitoring report have been cross-checked and reviewed as applicable against the PDMR and applied methodology.

The following list shows the default values used in this verification for GHG reduction calculation:

Table 4.1.1 Assessment of ex-ante determined parameters

Parameter	Description	Value applied	Assessment
EF _{grid,CM,y}	Combined Margin emission factor	0.602 tCO ₂ /kWh	Government of Myanmar has not published data on

			combined margin emission factor of its national grid. Hence, PP has used IFI Dataset of Default Grid Factors, version 3.2 published in April 2022 to apply combined margin grid emission factor for intermittent energy sources. This is the most recent data available at the time of submission of PDMR to the DOE for validation and found to be in line with approach defined in calculating CM using simple OM in Tool07.
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Table 4.1.2 Assessment of ex-post determined parameters

Table 14.12: Assessment of expected determined parameters							
Parameter	Description	Value applied	Assessment				
EG _{facility,y}	Quantity of net electricity generation supplied by the project plant to grid in year y	For 2019: 21,411 MWh 2020: 80,934 MWh 2021: 77,461 MWh 2022: 81,162 MWh	The project activity supplies electricity to two 230 kV substations namely Mann and Ann. There are one main and one back up meter located at 230kV feeder end to measure electricity export and import. Every month, site staff and substation representative take reading from main and back up export and import meters and record it in the form of 'Month-Year/meter record export' and 'Month-Year/meter record export' report. This is used to prepare 'total net send out units' report monthly by PP and issue a monthly invoice to EGPE by the PP for quantity of net electricity supplied. Details of main and back up meters installed at 230kV feeder is given below: <table><tr><th>Meter</th><th>Sr. No.</th></tr><tr><td>Mann main meter</td><td>XS.66083</td></tr></table>	Meter	Sr. No.	Mann main meter	XS.66083
Meter	Sr. No.						
Mann main meter	XS.66083						

			export and import	
			Mann back up meter export and import	XS.66084
			Ann main meter export and import	XS.66085
			Ann back up meter export and import	XS.66086
All the meters are 0.5s class and calibrated by EGPE. The calibration frequency is as per yearly. Calibration of these meters has been confirmed by checking calibration certificates/calibration certificates/				

In conclusion, GHG emission reductions provided in the project's GHG statement in the PDMR have been quantified correctly in accordance with the monitoring plan and applied methodology for this verification period.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

During the verification assessment, the evidence provided by the PPs was sufficient in both quantity and quality to support the determination of GHG emission reductions reported by the project. The threshold for materiality to the total GHG emission reductions reported was met for this project; materiality is a concept that the errors, omissions and misrepresentations could affect the GHG removals assertion and influence the intended users.

The source and nature of the evidence (external or internal, oral or documented) for the determination of GHG emission reductions or removals has been assessed and verified in above section 4.1 together with each calculation step, thus the reliability of the evidence is confirmed. The evidence provided to determine GHG removals reported in the Monitoring Report included values, notations, units and sources; the evidences has been cross-checked with supplied GHG emission reduction calculation spreadsheet^{ER/}.

The actual procedure for data generation and aggregation, recording, transfer and final transposition for quantification of GHG ER was also verified and found to be in compliance with the monitoring plan in the PDMR/PDMR/.

5 VALIDATION AND VERIFICATION OPINION

TÜV SÜD South Asia Private Limited has performed joint validation and verification of the emission reductions reported for the project activity “220 MWDC Minbu Solar Power Plant Project” in Myanmar, VCS Registration Reference No. 4030, for the period 27-September-2019 to 31-December-2022, with regard to the relevant requirements for VCS Version 4.4 and ISO 14064-3:2019.

The project proponent is GEP (Myanmar) Company Limited is responsible for:

1. The preparation of greenhouses gas emissions data and the reported greenhouse gas emission reductions from the project on the basis set out in the monitoring plan contained in the PDMR, version 5 dated 06-Mar-2025;
2. The development and maintenance of records and reporting procedures in accordance with that plan, including the calculation and determination of greenhouse gas emission reductions of the project.

Assumptions, limitations and methods included in the validation process were reasonable and the project is likely to achieve estimated GHG emission reduction or removals, explaining that actual results may vary since the estimates are based on assumptions that are subject to change. For verification, reasonable level of assurance was achieved on the quantity of GHG emission reductions achieved by the project during the verification period 27-September-2019 to 31-December-2022 as provided in the project’s GHG statement given in the PDMR version 5 dated 05-March-2025.

Validated GHG emission reductions in the period from 27-September-2019 to 26-September-2026, calendar year wise break down of emission reductions are given in the table below:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	48,581
Year 2	84,677
Year 3	132,578

Year 4	194,714
Year 5	210,221
Year6	208,725
Year7	207,241
Total estimated ERs	1,086,738
Total number of crediting years	7
Average annual ERs	155,248

For verification period from 27-September-2019 to 31-December-2022, calendar year wise break down of GHG emission reductions is given in the table below:

Year	Baseline emissions or removals (tCO₂e)	Project emissions or removals (tCO₂e)	Leakage emissions (tCO₂e)	Net GHG emission reductions or removals (tCO₂e)
27-September-2019 to 31-December-2019	12,890	0	0	12,890
01-January-2020 to 31-December-2020	48,722	0	0	48,722
01-January-2021 to 31-December-2021	46,632	0	0	46,632
01-January-2022 to 31-December-2022	48,860	0	0	48,860
Total	157,103	0	0	157,103

APPENDIX 1: ABBREVIATIONS

Abbreviations	Full Texts
AFOLU	Agriculture, forestry and other land use
BM	Build Margin
BE _y	Baseline Emissions
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
CM	Combined Margin
CO ₂	Carbon di oxide
CH ₄	Methane
ECD	Environmental Conservation Department
ER	Emission Reductions
EF	Emission Factor
EGPE	Electric Power Generation Enterprise
EMP	Environmental Management Plan
ESIA	Environmental Social Impact Assessment
EPC	Erection, Procurement and Commissioning
FAR	Forward Action Request
GHG	Greenhouse gas(es)
GS	Gold Standard
GCC	Global Carbon Council
kWh	Kilowatt hour
kV	Kilo Volt
MoM	Minutes of Meeting

MIC	Myanmar Investment Commission
MWDC	Megawatt of power with direct current
MWAC	Megawatt of power with alternating current
MWh	Megawatt hour
N ₂ O	Nitrous oxide
LSC	Local Stakeholder Consultation
L _y	Leakage emissions
OM	Operating margin
ODA	Overseas Development Assistance
PD	Project Description
PE _y	Project Emissions in a year
PDMR	Joint Project Description and Monitoring Report
PP	Project Proponent
PV	Photovoltaic
PPA	Power Purchase Agreement
SDG	Sustainable development goal
SSC	Small scale
SCADA	Supervisory control and data acquisition
TUV SUD	TUV SUD South Asia Private Limited
tCO ₂ e	Ton of Carbon dioxide equivalent
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard

APPENDIX 2: REFERENCE DOCUMENTS

/VCS/	1. VCS Standard Version 4.4 2. VCS Program Guide Version 4.3 3. VCS Program Definition Version 4.3
/ISO/	ISO 14064-2:2019 ISO 14064-3:2019
/Meth/	ACM0002 “Grid-connected electricity generation from renewable sources” Version 21
/Tool/	Tool01 “Tool for the demonstration and assessment of additionality” Version 7, Tool07 “Tool to calculate the emission factor for an electricity system” Version 7
/PDMR/	PDMR version 1 dated 13-Jan-2023 (Initial) PDMR, Version 5, dated 06-Mar-2025 (Final)
/ER/	ER calculations version 1 dated 13-Jan-2023 (Initial) ER calculations version 5 dated 06-Mar-2025 (Final)
/VVB Contract/	Validation and verification contract signed on 27/12/2022
/Insp/	1. List of personnel interviewed during on-site audit 2. Audit notes
/Declaration/	1. Confirmation from PP on not seeking environmental credits than VCUs 2. Confirmation from PP for not obtaining ODA
/Commissioning certificate/	Ministry of Electricity and Energy, determining of the commercial operation date dated 18/11/2019
/FS/	Feasibility study of solar power plant project in Minbu, Magway Region by Shiawatra University dated 02/07/2013, 04/10/2017
/ESIA/	Implementing solar power plant project in Minbu district, Magwe Region ESIA report by Myanmar Survey Research dated 20/10/2016
/MIC/	Myanmar Investment Commission permit for the investment in 220 MWDC solar power plant to GEP (Myanmar) Co. Ltd. dated 12/04/2017

/PPA/	<i>Power Purchase Agreement between Ministry of Electric Power Myanmar Electric Power Enterprise and GEP (Myanmar) Co. Ltd. for 220 MWDC dated 20/03/2016</i>
/EPC/	<i>Turnkey engineering, procurement and construction contract for 220MWDC solar power plant in Minbu, Myanmar with China Triumph International Engineering Company Ltd. dated 04/08/2016</i>
/WI/	<i>Work Instruction Energy meter export calculation and revenue meter monthly report WI-OPT-027</i>
/INV/	<i>Invoices issued to EGPE for sale of electricity for 2019, 2021, 2022</i>
/REG/	<i>The Electricity Law 2014, The Environmental Conservation Law 2012, The Forest Law , 1992, The Social Security Law, 2012, The minimum wage Law 2013</i>
/RC/	<i>Company registration for GEP (Myanmar) Co. Ltd.</i>
/LSC/	<i>MOM for public consultation dated 12/09/2016</i>
/Tech Spec/	1. Jettion Solar Multi-crystalline 156X156 module 2. Siemens transformer 50 MVA 3. SMA Sunny Central 1000CP XT inverter 4. CFAT DC 1000V combiner box
/Equipment list/	<i>Equipment list for 40MW Phase 1 at MSPP</i>
/Exemption/	<i>Letter issued by VERRA granting exemption from section 3.7.1 of VCS Standard 4.3 dated 14/12/2022</i>
/Biannual Report/	<i>1st 2nd 3rd 4th 5th, 6th, 7th, 8th, 9th and 10th biannual monitoring reports prepared by Myanmar Survey Research covering monitoring period from Jan 2018 to Dec 2022</i>
/Verra website/	<u>Verra Search Page</u>
/Calibration certificates/	<i>Calibration certificates for 230kV Mann main and back up meter and Ann main and back up meter</i>
/IFI/	<i>IFI dataset for default grid factors version 3.2 published in April 2022</i>
/Daily wage record/	<i>Labour charges form</i>
/UNFCCC-CDM/	<u>CDM: CDM Registry (unfccc.int)</u>

/GS/	GSF Registry (goldstandard.org)
/GCC/	Ho-e - GCC Projects Portal (globalcarboncouncil.com)
/CAR/	Climate Action Reserve - Climate Action Reserve : Climate Action Reserve
/NERL/	Technical paper published by NERL, referred regarding Residual Value.
/RES/	Renewable energy snapshot: Myanmar 香港特别行政区 Global law firm Norton Rose Fulbright).

APPENDIX 3: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS

CAR ID	CAR 1	Section no.	General Requirements	Date: 26/07/2023
Description of CAR				
PP is requested to submit PDMR using the latest available template version 4.2.				
Project participant response				Date: 10/08/2023
PP has submitted the PDMR template to the latest version 4.2				
Documentation provided by project participant				
	Changes in the PDMR	Section(s):	New version no 1.1	
	Changes in ER xls	Worksheet(s):	New version no:	
	Other:			
VVB assessment				Date: 15/09/2023
It is checked that PP has submitted PDMR using latest applicable template 4.2. Hence, closed.				

CAR ID	CAR 2	Section no.	1.1, 1.3, 1.9, 1.10, 1.14, 1.16, 1.17, 2.2, 2.3	Date: 26/07/2023
Description of CAR				

PP is requested to bring more clarity on the following points in PDMR according to VCS Standard 4.4.:

1. In section 1.1, the summary of total GHG emission reductions generated in the current monitoring period is not provided.
2. Section 1.3 does not clearly describe and justify how the project is eligible under the scope of the VCS program.
3. Section 1.10 does not describe project scale correctly. Also, it considers Year 11 in the total crediting period of 10 years which is incorrect. Crediting period selection is not in accordance with VCS standard.
4. Currently, PDMR refers to certain certificates & permits in 1.10. PP is required to list all national, regional and local laws, statutes and regulatory framework in Myanmar that are relevant to the project activity and provide an assurance that the project is complying with these laws and regulations and where relevant, demonstrate how the compliance is achieved.
5. PP is requested to provide a declaration that the project does not generate other forms of GHG-environmental credits such as RECs or other and that the VCUs will not be issued as GHG allowances or other types of GHG credits under and emission trading program. If the VCUs will be used in the context of Article 6 mechanisms and international Paris agreement related programs, then PP needs to clearly mention this in the PDMR or else include this in the above declaration.
6. PP is requested to explain the PDMR how it has ensured that the VCUs generated by the project activity will not be double claimed in Scope 3 emissions statements made by utility company on generation of purchased electricity that is sold to end users.
7. In 1.17,
 - PDMR does not describe 7.b contributions over project lifetime correctly.
 - Indicator selected for target 8.5 is not clear. Contributions over project lifetime for this target are not clear.
 - For 9.1.1, current & over project lifetime contributions does not justify the target & indicator's expected contributions.
 - Indicator selected for target 13 is not clearly described. Contributions over project lifetime are incorrect.
8. PP has not selected project crediting period in line with the requirements of standard.
9. PDMR lacks information ongoing communication mechanism established by PP, any input received during local stakeholder consultation and through ongoing communications, how it has been taken into account and whether it has resulted in change in project design.

10. Please provide EIA study report referred in 2.3.

11. Section 2.4 is missing in the PDMR.

Project participant response

Date: 10/08/2023

1. Section 1.1, PP has provided summary of total GHG emission reductions generated in the current monitoring period.
2. Section 1.3. PP has an additional describe the project to justify how is eligible under the scope of the VCS program.
3. Section 1.10. PP has revised the project scale from a large project to a project. And as we start crediting period on 27-09-2023 and has chosen 7 years renewal crediting period.
4. Section 1.14. PP has provided the list of national, regional, and local laws, statutes, and regulatory frameworks in Myanmar that are relevant to the project activity and provide an assurance that the project is complying with these laws and regulations where relevant.
5. Section 1.16.1 PP has provided a clear statement to declare that the project does not generate other forms of GHG-environmental credits such as RECs or others and that the VCUs will not be issued as GHG allowances or other types of GHG credits under emission trading program.
6. Section 1.16.3 PP explained in the PDMR how it has ensured that the VCUs generated by the project activity will not be double claimed in Scope 3 emissions statements.
7. Section 1.17 PP has provided a new description of the current & project lifetime of SDG activities monitoring in 7.2, 8, 13
8. Section 1.9 PP has provided the project crediting period in PDMR with the standard's requirements.
9. Section 2.2 has provided the information and input of stakeholder consultation activities reference in the ESIA Report 2016
10. Section 2.3 PP has provided the ESIA report which can us as an EIA report
11. Section 2.4 PP has provided the missing section 2.4 by the filled status of the public comments period by VERRA

Documentation provided by project participant

	Changes in the PDMR	Section(s) 1.1, 1.3, 1.9, 1.10, 1.14, 1.16, 1.17, 2, 2.3 and 2.4	New version no:1.1
	Changes in ER xls	Worksheet(s):	New version no:
	Other:	3 rd EIA report, 8 rd -10 rd EIA report,	
VVB assessment			15/09/2023
Team has checked response provided by and changes made in the PDMR & ER and found to be accepted. Hence, closed.			

CAR ID	CAR 3	Section no.	3.2, 4.1, 4.2, 4.4, 5.1, 5.2, 6, Appendix 1	Date: 26/07/2023
Description of CAR				

PP is requested to bring more clarity on the following points in PDMR according to VCS Standard 4.4.:

1. Section 3.2 does not clearly describe how the project meets each of the applicability conditions of ACM0002 & applied tools as given in respective applicability sections.
2. PP needs to explain in the PDMR how it has estimated emission factor $EF_{grid,CM,y}$ in accordance with the Tool to calculate emission factor for an electricity system Version 7.
3. PDMR is not clear on assumptions/data used in calculation of baseline, project & leakage emissions or exclusion of any emission sources if any under these 3 emission scopes.
4. Section 4.4 includes Year 11, whereas crediting period chosen is of 10 years.
5. Data and parameters not monitored are inconsistent with ACM0002 Version 21.
6. Data and parameters monitored $EG_{m,y}$, EG_y , $EG_{k,y}$ and $EG_{n,h}$ is not found in accordance with ACM0002 Version 21.
7. PP has not provided monitoring plan for applicable parameters as per para 73 of ACM0002 Version 21.
8. The monitoring plan section misses procedures used for internal auditing, procedures used for handling any internal auditing performed and any non-conformance identified, policies used for oversight and accountability of monitoring activities, the organizational structure and competencies of the personnel that carry out monitoring activities, methods used for storing, aggregating, collating and reporting data and parameters.
9. Section 7 is missing in the PDMR.
10. The organization structure included in Appendix 1 is non-readable.

Project participant response

Date: 10/08/2023

1. Section 3.2 PP clearly describes how the project meets the applicability conditions of ACM0002 & applied tools as given in respective applicability sections.
2. PP has revised PDMR in how it has estimated emission factor $EF_{grid,CM,y}$ in accordance with the Tool to calculate emission factor for an electricity system Version 7.
3. PP has revised PDMR to clarify assumptions/data used in calculation of baseline, project & leakage emissions .
4. 4.4 PP starts crediting period on 27-09-2023, which overlaps another nine months of the year 2029 to complete ten year crediting period.
5. Exclude section data and parameters not monitored.
6. PP has revised the Data and parameters monitored section.
7. PP has revised the monitoring plan for applicable parameters per para 73 of ACM0002 Version 21.
8. PP has revised the monitoring plan procedures as required in PDMR.
9. PP has added section 7 which missing in PDMR.
10. Organization structure has been revised in Appendix 1

Documentation provided by project participant

	Changes in the PDMR	Section(s): 3.2, 4.1, 4.2, 4.4, 5.1, 5.2, 6, Appendix 1	New version no: 1.1
	Changes in ER xls	Worksheet(s):	New version no:
	Other:		

VVB assessment
Date: 15/09/2023

Team has checked response provided by and changes made in the PDMR & ER and found to be accepted. Hence, closed.

CL ID	CL 1	Section no.	1.1	Date: 26/07/2023
Description of CL				
The feasibility report presented to Ministry of Electric Power describes phase wise commissioning of project as Phase 1- 50MWp, Phase 2- 85MWp and Phase 3- 85MWp. The information furnished in the PDMR is consistent with this. PP is requested to clarify this.				

Project participant response			Date: 10/08/2023
Please see the attached file for relevant feasibility report which is in line with the project description (The project consists of phase 1 of 50MW, phase 2 of 50MW, phase 3 of 50 MW, and phase 4 of 70 MW) in section 1.1. For further evidence, please refer to Revise Feasibility Study (page 8)			
Documentation provided by project participant			
	Changes in the PD MR	Section(s):	New version no: 1.1
	Changes in ER xls	Worksheet(s):	New version no:
	Other:	Attached are the feasibility report and PPA	
VVB assessment			Date: 15/09/2023
PDMR is consistent with PDMR. Hence, accepted. CL is closed.			

CL ID	CL 2	Section no.	1.7	Date: 26/07/2023
Description of CL				
PP is requested to provide evidence establishing project ownership for 220MWp as per 3.7.1 of VCS Standard 4.4 and clearly demonstrate the same in PDMR.				
Project participant response				Date: 10/08/2023
PP has provided evidence of establishing the project ownership for 220 MWp in PPA page 1.				
Documentation provided by project participant				
	Changes in the PD MR	Section(s): 1.7	New version no: 1.1	
	Changes in ER xls	Worksheet(s):	New version no:	
	Other:	PPA		
VVB assessment				Date: 15/09/2023
Evidence is checked and found to be acceptable. Hence, CL is closed.				

CL ID	CL 3	Section no.	1.1, 1.8	Date: 26/07/2023
Description of CL				

The Letter from Ministry of Electricity and Energy dated 15/11/2019 describes that 40MW capacity started commercial operation on 27/09/2019 in Phase 1. Whereas the PDMR describes 50MWp has begun operation in Phase 1. PP is requested to clarify this.			
Project participant response			Date: 10/08/2023
The company has an installed capacity of 50MW _{DC} converted to 40 MW _{AC} . PDMR has been revised to state the unit used clearly.			
Documentation provided by project participant			
	Changes in the PDMR	Section(s): 1.1	New version no: 1.1
	Changes in ER xls	Worksheet(s):	New version no:
	Other:	Revise feasibility study	
VVB assessment			Date: 15/09/2023
Revised PDMR has been checked and found to be in order. Hence, accepted.			

CL ID	CL 4	Section no.	2.1	Date: 26/07/2023
Description of CL				
3 rd environmental monitoring report of Phase 1 50MWp conducted by MSR states environmental & social impacts due to project activity. Description given 2.1 is inconsistent with this monitoring report. PP is requested clarify this and make 2.1 consistent with the findings of this report.				
Project participant response				Date: 10/08/2023
Section 2.1 is revised to be in line with 3 rd environmental monitoring report.				
Documentation provided by project participant				
	Changes in the PDMR	Section(s): 2.1	New version no: 1.1	
	Changes in ER xls	Worksheet(s):	New version no:	
	Other:	Attached 3 rd the environmental monitoring report for reference		
VVB assessment				Date: 15/09/2023
Revised PDMR has been checked and found to be in order. Hence, accepted.				

CL ID	CL 5	Section no.	3.5	Date: 26/07/2023
Description of CL				
1. PP is requested to provide evidence to substantiate that the project is first solar power plant in Myanmar. If so, then why PP has not applied guidelines on additionality of first-of-its-kind project activities. 2. PP has considered only 50MWp capacity for carrying out investment analysis whereas the project is of 220MWp. 3. PP is needed to clearly mention all the input data used in investment analysis, sources of each data and to furnish supporting documents for electricity tariff, degradation rate, project life, capital cost, O&M cost, MMK/USD exchange rate, tax rate, depreciation and any other data that has been used in calculating IRR in PDMR. 4. PP is to explain how lending rate has been selected as a benchmark according to para 38 of tool for the demonstration and assessment of additionality and how the input data was available at the time of decision making. 5. PP has missed to discuss step 1b in the PDMR. 6. PP is requested to clarify basis of selecting tariff, investment & operating cost for performing sensitivity analysis and how the range of variations is reasonable in the project context. 7. PP is requested to clarify in the PDMR at what range of variations the IRR crosses benchmark for the selected variables and probability of the occurrence of this scenario.				
Project participant response				Date: 10/08/2023
For findings 1 to 7, PP has selected to apply first of its kind for demonstrating additionality. Hence, earlier argument of additionality has been modified.				
Documentation provided by project participant				
	Changes in the PDMR	Section(s): 3.5	New version no:	
	Changes in ER xls	Worksheet(s):	New version no:	
	Other:			
VVB assessment				Date: 15/09/2023
Response provided by the PP has been checked and accepted. Hence, CL is closed.				

FAR ID	FAR 1	Section no.		Date: 20/09/2023
Description of CL				
VVB carrying out 2 nd verification should check implementation and operation of remaining capacity of project activity as planned.				
Project participant response				Date:
Documentation provided by project participant				
	Changes in the PDMR	Section(s):		New version no:
	Changes in ER xls	Worksheet(s):		New version no:
	Other:	Revise feasibility study		
VVB assessment				Date:

APPENDIX 4:COMPETENCY CERTIFICATES

T ◆ CERTIFICATE ◆ 認証証書 ◆ CERTIFICADO ◆ CERTIFICAT



CERTIFICATE OF APPOINTMENT

Ms. Shruti Kudtarkar fulfills the requirements of the Certification Body 'Environment and Energy' of TÜV SÜD South Asia Pvt Ltd to participate in audits.

Qualification applicable to					
Standard	CDM	GS	VCS	ISO-14064-1, 2	Other- GCC, PAS 2060
	☒	☒	☒	☒	☒

Qualified as						
Status	Validator	Verifier	ATL	Technical Reviewer	Financial Expert	Technical Expert
	☒	☒	☒	☒	☒	☒
TA- *A	CDM - 1.2, 13.1 SO 14064-1 – (11,13) ISO 14064-2 - (1,13)					
TA- **UA	-					

Country Expertise						
Region	1 Americas ☐	2 ASEAN, South Asia, Middle East & Africa ☒	3 Central & Eastern Europe ☐	4 Germany ☐	5 North Asia ☐	6 Western Europe ☐
Further Countries	India, Malaysia, Sri Lanka, Indonesia, Paraguay, Myanmar					

Technical Area (TA) / Scope
TA- *Accredited CDM/GCC- 1.2_Renewables, 13.1_ Waste handling and Disposal. ISO 14064-1- Waste handling and disposal, General. ISO 14064-2- Energy industries (renewable), Energy demand, Waste handling and disposal VCS - Energy Industries (renewable), Waste Handling and Disposal

This appointment is valid until 25/10/2024 and is bound by internal requirements of the Certification Body 'Environment and Energy' of TÜV SÜD South Asia Pvt Ltd.

In case of loss of validity of this certificate as per result of an assessment according to internal procedures or due to any other reason, it will be properly communicated to you.
Your Certificate has the internal reference no. CB-IND-CCP-0022/004.

Date	Signature
04/03/2024	 Deepankar Chowdhury

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ZERTIFIKAT



CERTIFICATE OF APPOINTMENT

Mr. Srinivasan Selvaraj fulfills the requirements of the Certification Body 'Environment and Energy' of TÜV SÜD South Asia Pvt Ltd to participate in audits.

Qualification Details

Standard	Sectoral Scope	Technical Area	Technical Area Description	Role
CDM/GCC/GS	SS1, SS3, SS5, SS 13	1.1, 1.2, 3.1, 5.1, 13.1, 13.2	1.1 Thermal Energy Generation 1.2 Renewables 3.1 Energy Demand 5.1 Chemical Industry 13.1 Solid waste and wastewater 13.2 Manure	VAL/VER/ATL/TR/TE
VCS	SS1, SS3, SS5, SS 13	1, 3, 5, 13	1 Energy (renewable/non-renewable) 3 Energy demand 5 Chemical industry 13 Waste handling and disposal	VAL/VER/ATL/TR/TE

Country Expertise: India

Certificate Number: CB-IND-CCP-0067/001

Issued on: 26/11/2024

Valid until: 25/11/2025

Version	Date	Reason for Revision
01	26/11/2024	Initial appointment

Legend:
VAL - Validator, VER - Verifier, ATL - Audit Team Leader, TR - Technical Reviewer, TE - Technical Expert, FE - Financial Expert, CE - Country Expert, EE - Environment Expert, SE - Social Expert
CDM - Clean Development Mechanism, VCS - Verified Carbon Standard, GS4GG - Gold Standard for Global Goals, GCC - Global Carbon Council, PPRS - Plastic Pollution Reduction Standard

This appointment is bound by internal requirements of the Certification Body 'Environment and Energy' of TÜV SÜD South Asia Pvt Ltd. In case of loss of validity of this certificate as per result of an assessment according to internal procedures or due to any other reason, it will be properly communicated to you.


Issued By
Quality Manager

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CERTIFICATE OF APPOINTMENT

Ms. Priyanka Mukherjee fulfills the requirements of the Certification Body 'Environment and Energy' of TUV SUD South Asia Pvt Ltd to participate in audits.

Qualification Details

Standard	Sectoral Scope	Technical Area	Technical Area Description	Role
CDM/GCC	SS1,SS3, SS13	1.2, 3.1, 13.1	1.2 Renewables 3.1 Energy Demand 13.1 Solid waste and wastewater	VAL/VER VAL/VER VAL/VER
GS	GS4GG Energy & Waste	1.2, 3.1, 13.1	1.2 Renewables 3.1 Energy Demand 13.1 Solid waste and wastewater	VAL/VER VAL/VER VAL/VER
VCS	-	1, 3, 13	1_Energy Industries (renewables) 3_Energy Demand 13_Waste Handling and Disposal	VAL/VER VAL/VER VAL/VER

Country Expertise: India

Certificate Number: CB-IND-CCP-0044/001

Issued on: 30/07/2024

Valid until: 29/07/2025

Version	Date	Reason for Revision
01	30/07/2024	Initial appointment

Legend:
VAL - Validator, VER - Verifier, ATL - Audit Team Leader, TR- Technical Reviewer, TE - Technical Expert, FE - Financial Expert, CE - Country Expert, EE - Environment Expert, SE - Social Expert
CDM - Clean Development Mechanism, VCS - Verified Carbon Standard, GS4GG - Gold Standard for Global Goals, GCC - Global Carbon Council, PPRS - Plastic Pollution Reduction Standard

This appointment is bound by internal requirements of the Certification Body 'Environment and Energy' of TUV SUD South Asia Pvt Ltd. In case of loss of validity of this certificate as per result of an assessment according to internal procedures or due to any other reason, it will be properly communicated to you.


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CERTIFIKAT



CERTIFICATE OF APPOINTMENT

Mr. Vivekanand Date fulfills the requirements of the Certification Body 'Environment and Energy' of TUV SUD South Asia Pvt Ltd to participate in audits.

Qualification Details

Standard	Sectoral Scope	Technical Area	Technical Area Description	Role
CDM, GS, VCS and GCC	-	-	-	FE

Country Expertise: NA

Certificate Number: CB-IND-CCP-0041/01

Issued on: 06/06/2024

Valid until: 31/05/2026

Version	Date	Reason for Revision
01	06/06/2024	Initial appointment

Legend:
VAL - Validator, VER - Verifier, ATL - Audit Team Leader, TR- Technical Reviewer, TE - Technical Expert, FE - Financial Expert
CDM - Clean Development Mechanism, VCS - Verified Carbon Standard, GS4GG - Global Standard for Global Goals, GCC - Global Carbon Council, PCX - Plastic Credit Exchange

This appointment is bound by internal requirements of the Certification Body 'Environment and Energy' of TUV SUD South Asia Pvt Ltd. In case of loss of validity of this certificate as per result of an assessment according to internal procedures or due to any other reason, it will be properly communicated to you.


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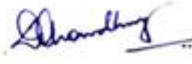
CERTIFICATE OF APPOINTMENT

Mr. Sudheendra, K fulfills the requirements of the Certification Body 'Environment and Energy' of TÜV SÜD South Asia Pvt Ltd to participate in audits.

Qualification applicable to						
Standard	CDM	GS	VCS	ISO-14064-1, 2	Other-GCC	
	☒	☒	☒	☒	☒	
Qualified as						
Status	Validator	Verifier	ATL	Technical Reviewer	Financial Expert	Technical Expert
	☐	☐	☐	☒	☐	☒
TA- *A	CDM / GCC - 1.1, 1.2, ISO 14064-1 - 1 ISO 14064-2 - 1 VCS - 1					
TA- **UA	-					
Country Expertise						
Region	1 Americas ☐	2 ASEAN, South Asia, Middle East & Africa ☒	3 Central & Eastern Europe ☐	4 Germany ☐	5 North Asia ☐	6 Western Europe ☐
Countries	India					
Technical Area (TA) / Scope						
TA- *Accredited CDM / GCC - 1.1_Thermal, 1.2_Renewable ISO 14064 -1 - Power Generation and Electric Power Transactions ISO 14064-2 - Energy industries (renewable/non-renewable sources) VCS - Energy industries (renewable/non-renewable sources)						

This appointment is valid until 31/07/2025 and is bound by internal requirements of the Certification Body 'Environment and Energy' of TÜV SÜD South Asia Pvt Ltd.

In case of loss of validity of this certificate as per result of an assessment according to internal procedures or due to any other reason, it will be properly communicated to you.
Your Certificate has the internal reference no. CB-IND-CCP-0104/006.

Date	Signature
20/05/2024	 Deepankar Chowdhury

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